Ti Grade 6Al-4V ELI Grade 233 | UNS# R56401

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Weight Calculator

Ti Grade 6Al-4V ELI Grade 23 is produced at a higher purity than Titanium 6Al-4V Grade 5. Though this alloy is mainly recognized as the best dental and medical Titanium grade, it is also widely used in applications when there is a need for toughness, high strength, light weight, and good corrosion resistance.

Because of these unique biocompatibility characteristics, Titanium 6Al-4v ELI Grade 23 is also used in bio-medical applications, implants, and similar surgical procedures and situations.

Titanium 6Al-4V ELI Grade 23 Specifications:

Titanium 6Al-4V ELI Grade 23 Round Bar

- AMS 2631, AMS 4930, AMS 6932, EN 10204 3.1, MIL-T-9047
- ASTM B348 Gr 23, ASTM F136

- ASTM F136, ASTM B348 Gr 23

Titanium 6Al-4V ELI Grade 23 Plate

- AMS 4907
- ASTM F136, ASTM B265 Gr 23

Titanium 6Al-4V ELI Grade 23 Sheet

- AMS 4907, AMS-T-9046
- ASTM B265 Gr 23, ASTM F136

Titanium 6Al-4V ELI Grade 23 Applications:

- Aircraft Applications
- Bone Fixation and Similar Devices
- Cryogenic and High Pressure Vessels
- Ligature Clips
- Marine and Salt Water Applications
- Orthopedic Cables, Pins, and Screws
- Orthodontic Appliances
- Surgical Implants, Joint Replacements and Staples

DISCLAIMER: Alloy data sheets are for reference only, not for design purposes.

CHEMICAL COMPOSITION				
Ti 88.09 – 91%	Al 5.5 – 6.5%%	V 3.5 – 4.5%	Fe	C

Density	Electrical conductivity (% IACS)	Thermal conductivity (BTU-in/hr-ft²-°F)	Specific heat capacity	Thermal expansion (1/°F)	Melting Point (°F)
0.160	0.97	46.5	560	0.00000478	3200

MECHANICAL PROPERTIES		
PROPERTY	MINIMUM	MAXIMUM
Yield Strength Range ($Rp_{0.2\ Offset}$) (ksi)	120	160
Tensile Strength Range (R_m) (ksi)	125	174
Elongation (AL=5 x Ø)	10.0%	18.0%
Brinell Hardness (H)	326	379
Modulus of Elasticity (ksi) (E): 16500		
Machineability General Index: 21%		